

FathomNet 2025: Multi-Scale Context and Taxonomic Distance Learning for Marine Species Classification

Final Project Report

Author: Dawood Aijaz

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Abstract. The FathomNet 2025 competition studies fine-grained recognition of marine organisms from underwater imagery. Each annotated region of interest (ROI) must be assigned a taxonomic label, and performance is measured with a distance matrix that penalizes distant biological mistakes more heavily than nearby ones. In this report, we present a multi-scale, taxonomy-aware classification system designed specifically for that evaluation setting. The method combines ROI crops with wider environmental context at 3x, 5x, and full-image scales, processes each scale with a dedicated ConvNeXtV2 encoder, fuses the resulting features, and predicts the complete taxonomic path from kingdom to species with hierarchical classification heads. To better align training with the leaderboard metric, we replace a purely flat species loss with a taxonomic distance-aware objective and additionally examine confidence-aware fallback as an auxiliary hierarchy-aware decision rule for uncertain cases. Across a series of ablations, multi-scale context substantially improves over an ROI-only baseline, and the final 4-scale model with taxonomic loss achieves a private leaderboard score of 1.94, improving on the baseline by about 27 percent. Saved evaluation reports and confusion matrices further show that the model preserves coarse taxonomic structure well even when leaf-level species prediction remains difficult.

1 Introduction

1.1 Motivation

Automated recognition of marine organisms is an important enabling technology for ocean science (Katija et al. 2022). Modern remotely operated vehicles and long-duration camera systems collect far more imagery than marine biologists can annotate by hand, and many organisms appear only as small, low-contrast, partially occluded regions embedded in cluttered underwater scenes. A practical recognition model must therefore do more than identify local texture; it must also reason about context, uncertainty, and biological structure. These requirements make FathomNet an instructive benchmark for studying classification methods that go beyond ordinary flat label prediction.

1.2 FathomNet 2025 competition description

The FathomNet 2025 competition provides underwater images together with bounding boxes around candidate organisms (Kaggle 2025). Each ROI is associated with a label drawn from a seven-level taxonomy running from kingdom to species. Unlike standard image-classification benchmarks, the com-

petition does not evaluate predictions with simple top-1 accuracy. Instead, it uses a taxonomic distance matrix derived from the biological hierarchy. Predicting the exact species yields zero penalty, predicting a nearby genus or family incurs a small penalty, and confusing distant branches of the tree incurs a much larger one. This metric encourages models to make biologically reasonable errors when exact recognition is difficult.

1.3 Objective and scope

The central objective of this work is to design a classifier whose architecture, loss, and inference rule are all aligned with that hierarchy-aware evaluation.

1.3.1 Objectives

- Build a multi-scale visual representation that combines fine ROI detail with surrounding habitat context.
- Predict the full taxonomic path rather than only the final species label.
- Incorporate the taxonomic distance matrix directly into training so that the model prefers nearby mistakes over distant ones.
- Investigate whether a confidence-aware fallback mechanism can improve robustness for uncertain

cases.

1.3.2 Scope

- We use only the competition data and taxonomy metadata supplied with the benchmark.
- The work focuses on supervised ROI classification rather than object detection or segmentation.
- Most experiments are designed for reproducible single-node, mixed-precision training with pre-trained ConvNeXt backbones.
- Evaluation is reported primarily with public and private leaderboard taxonomic distance scores, where lower is better.

2 Background and Related Work

2.1 ConvNeXt-style visual backbones

Transfer learning remains a strong starting point for fine-grained visual classification. ConvNeXt and ConvNeXtV2 offer modern convolutional backbones with strong ImageNet pretraining, efficient scaling behavior, and robust performance on natural-image tasks (Liu et al. 2022; Woo et al. 2023). They are well suited to FathomNet because ROIs often contain subtle texture and shape cues that benefit from high-quality convolutional feature extraction, while still being practical enough to fine-tune across multiple scales. In this setting, these backbones serve as the core feature extractors for both the tight ROI crop and the wider context crops.

2.2 Hierarchical and cost-sensitive classification

The competition metric makes this task a natural example of cost-sensitive classification. Standard cross-entropy assumes that every mistake is equally bad, but biological taxonomy is explicitly structured: confusing two closely related organisms should cost less than confusing organisms from different phyla. Prior work on hierarchical classification, hierarchy-aware deep networks, and cost-sensitive learning shows that embedding class relationships into either the loss or the decision rule can improve both robustness and interpretability (Bertinetto et al. 2020; Elkan 2001; Yan et al. 2015). My approach follows that principle by weighting species-level training with a taxonomic distance term rather than optimizing only for flat accuracy.

2.3 Multi-scale context and uncertainty-aware prediction

Marine organisms are often difficult to identify from the cropped organism alone. Local background, neighboring organisms, substrate type, and scene composition can all provide useful cues. This motivates a multi-scale design in which the model sees both the ROI and progressively larger windows around it, consistent with prior work showing that recognition can benefit from combining fine local evidence with broader contextual structure (Fu, Zheng, and Mei 2017; Lin et al. 2017). At the same time, the distance-matrix metric rewards conservative decisions when the model is uncertain. Confidence-aware fallback is therefore a natural complement to multi-scale modeling: if the model is not confident about the species, it can retreat to a taxonomically safer alternative derived from higher-level predictions, an idea related to hierarchical hedging, calibration, and selective prediction (Deng et al. 2012; Guo et al. 2017; Geifman and El-Yaniv 2019).

3 Data and Problem Formulation

3.1 Dataset and annotation structure

The training data contain 8,981 underwater images and 23,699 ROI annotations spanning 79 competition categories. Each annotation is defined by a COCO-style bounding box and linked to a taxonomic concept. In addition to the ROI coordinates, the dataset includes a taxonomy table that maps each concept through seven biological ranks: kingdom, phylum, class, order, family, genus, and species. A precomputed distance matrix derived from that hierarchy provides the competition score.

To support multi-scale learning, we generate several crops for each annotation. The tight 1x crop preserves fine morphology, the 3x and 5x crops include progressively more environmental context, and a full-image scale captures the global scene. All scales are resized to a common spatial resolution before being passed to the model.

Operationally, the pre-cropping stage uses deterministic expansions around the ROI. The 1x crop keeps a tight view with about 10 percent padding around the box, the 3x crop uses a 200 percent margin, the 5x crop uses a 400 percent margin, and the full-image view retains scene-level context such as substrate, lighting, and neighboring organisms. Precomputing these crops reduces training overhead and makes the input pipeline consistent

Multi-scale crop generation from each annotated ROI

Each annotation produces four aligned views that trade off local detail and scene context.

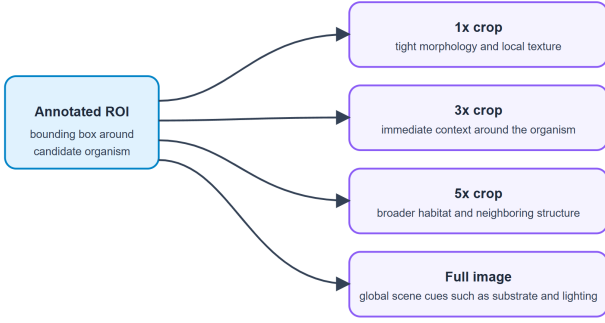


Figure 1: Multi-scale crop generation from an annotated ROI.

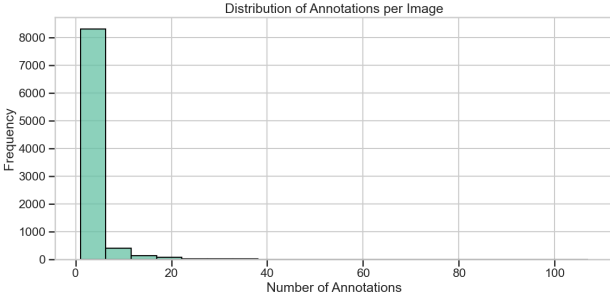


Figure 2: Distribution of annotations per image.

across epochs.

Figure 1 summarizes the four crop views derived from each annotated ROI.

3.2 Exploratory data analysis

Several statistics of the training set guided the final design:

- The dataset contains 23,699 ROIs over 8,981 images, or 2.64 annotations per image on average.
- The median image has only one annotation, but the most crowded image contains 107, so contextual modeling is still valuable.
- Bounding-box area ranges from 63 pixels to 2,075,604 pixels, with a mean of 39,296 and a median of 4,950, indicating many small targets together with a few very large specimens.
- ROI aspect ratios are usually close to square (mean 1.16, median 1.08) but can become elongated, reaching about 8.83 in the most extreme cases.

These observations justify three design choices: strong image augmentation, standardized 224 x 224 inputs for all scales, and explicit use of surrounding context for small or ambiguous organisms.

Figure 2 shows how annotation counts are distributed across images in the training set.

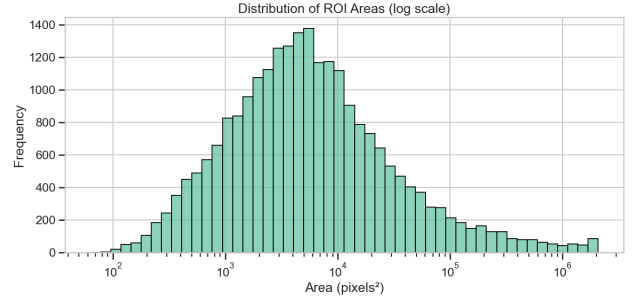


Figure 3: Distribution of ROI areas.

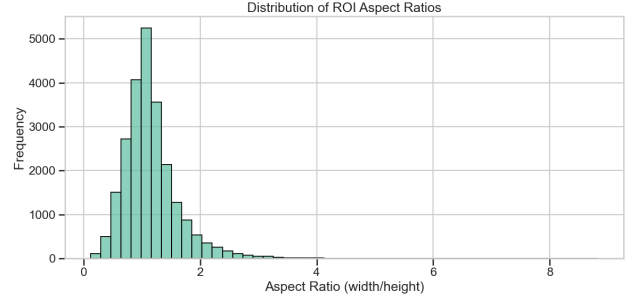


Figure 4: Distribution of ROI aspect ratios.

Figure 3 summarizes the scale variation of ROI sizes, which is one of the main reasons multi-scale crops are needed.

Figure 4 highlights the shape variability of the annotated boxes.

3.3 Label distribution and taxonomic structure

The class distribution at the leaf level is nearly uniform: each of the 79 categories contributes approximately 299 to 300 annotations. This is useful because it means the main difficulty is not severe leaf-class imbalance, but rather the combination of subtle visual differences and a structured taxonomic output space. The hierarchy itself remains nontrivial, with 1 kingdom, 7 phyla, 22 classes, 42 orders, 65 families, 74 genera, and 79 species represented in the taxonomy table. During learning, we also reserve an additional unknown bucket at each rank so that incomplete or ambiguous labels can be handled without breaking the hierarchical output structure. Because the evaluation metric is built from this hierarchy, the task is best formulated not as ordinary flat classification, but as structured prediction over a biological tree.

Figure 5 makes the near-uniform leaf-class distribution visually explicit.

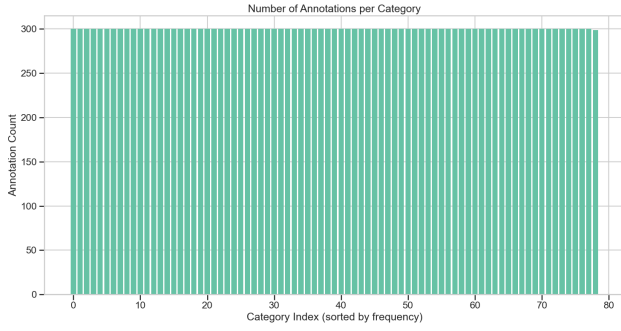


Figure 5: Annotation counts per category.

End-to-end classification pipeline

The system combines multi-scale visual encoding with hierarchical prediction and taxonomy-aware decision making.

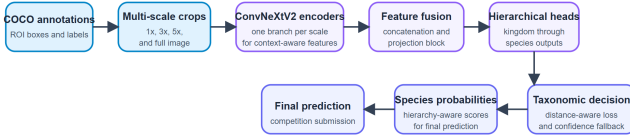


Figure 6: Overall classification pipeline.

4 Methods

4.1 Overall pipeline

Our pipeline begins by reading the COCO annotations, generating or loading pre-cropped views of each ROI at multiple scales, and splitting the data into stratified train, validation, and evaluation partitions. Each scale is passed through its own pretrained backbone, the scale-specific features are concatenated and projected into a shared representation, and that shared representation is decoded into seven taxonomic predictions. During training, the model is optimized with weighted hierarchical losses, including a taxonomic distance-aware objective at the species level. During inference, the predicted species can be replaced by a safer fallback when the species confidence is too low.

The complete flow is illustrated in Figure 6.

4.2 Multi-scale ConvNeXt backbone and feature fusion

The core visual encoder is a multi-branch ConvNeXtV2 architecture. Each input scale is processed by its own backbone so that the ROI branch can focus on organism morphology while the wider branches learn contextual scene cues. In the main taxonomic-loss configuration, the active scales are encoded with ConvNeXtV2-Base feature extractors, and the best leaderboard submission extends the same design from three scales to four by adding the full-image branch. Regardless of the exact scale set, every branch produces a pooled feature vector, and the vectors are concatenated into a single multi-

Hierarchical prediction heads with top-down conditioning

Each taxonomic head receives the shared fused feature and the probability output from the previous level.

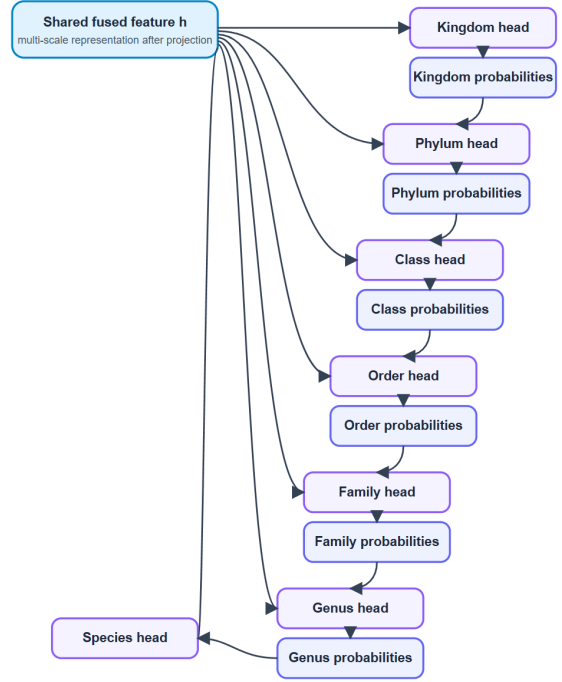


Figure 7: Hierarchical prediction heads.

scale descriptor. This design choice follows the broader intuition from multi-scale representation learning and fine-grained recognition that different spatial extents capture complementary discriminative signals (Fu, Zheng, and Mei 2017; Lin et al. 2017).

This fused representation is passed through a compact projection block composed of layer normalization, a linear layer, GELU activation, and dropout. The result is a shared feature space used by the downstream taxonomic heads. We favored concatenation over more elaborate fusion because it was simple, stable, and competitive in practice, and the heavier attention-based variants explored in this project ultimately serve as ablations rather than the core reported method.

The hierarchical conditioning structure is illustrated in Figure 7.

4.3 Hierarchical prediction heads and taxonomic loss

Instead of predicting only the final class, the model predicts all seven taxonomic levels. The top-level kingdom logits are produced directly from the shared features, and each lower-level head conditions on both the shared representation and the predicted probability distribution from the level above. Writing this cascading hierarchy explicitly, with $\mathbf{z}$ denoting the shared fused representation,

gives

$$\begin{aligned}
\mathbf{p}^{(\text{kingdom})} &= h_{\text{kingdom}}(\mathbf{z}), \\
\mathbf{p}^{(\text{phylum})} &= h_{\text{phylum}}(\mathbf{z}, \mathbf{p}^{(\text{kingdom})}), \\
\mathbf{p}^{(\text{class})} &= h_{\text{class}}(\mathbf{z}, \mathbf{p}^{(\text{phylum})}), \\
\mathbf{p}^{(\text{order})} &= h_{\text{order}}(\mathbf{z}, \mathbf{p}^{(\text{class})}), \\
\mathbf{p}^{(\text{family})} &= h_{\text{family}}(\mathbf{z}, \mathbf{p}^{(\text{order})}), \\
\mathbf{p}^{(\text{genus})} &= h_{\text{genus}}(\mathbf{z}, \mathbf{p}^{(\text{family})}), \\
\mathbf{p}^{(\text{species})} &= h_{\text{species}}(\mathbf{z}, \mathbf{p}^{(\text{genus})}).
\end{aligned}$$

Training uses a weighted sum of cross-entropy losses across levels, with larger weights assigned to the lower levels because species prediction is the quantity scored on the leaderboard. In the main configuration, the level weights increase from 0.5, 0.75, 1.0, 1.25, 1.5, 2.0 to 2.5 from kingdom through species. At the species level, we replace plain cross-entropy with a taxonomic distance-aware loss:

$$L_{\text{species}} = (1 - \alpha) \text{CE} + \alpha \sum_i p_i D(i, y)$$

where p_i is the predicted probability of species i , y is the ground-truth species, and $D(i, y)$ is the distance from the taxonomy matrix. In the best-performing setup, $\alpha = 0.3$. This term penalizes spreading probability mass toward distant species and encourages the model to remain biologically close when uncertain. The implementation also supports taxonomic label smoothing, although the distance-aware loss is the central mechanism in the strongest results. Conceptually, this objective is aligned with hierarchy-aware deep classification, cost-sensitive decision making, and label-smoothing-style regularization, all of which aim to distinguish severe mistakes from biologically nearby ones (Bertinetto et al. 2020; Elkan 2001; Yan et al. 2015; Szegedy et al. 2016).

4.4 Confidence-aware fallback at inference time

The broader system design also includes a confidence-aware fallback rule. If the model is sufficiently confident in the predicted species, that species is emitted directly. Otherwise, the system backs off through higher taxonomic levels, beginning at genus and moving upward through family, order, class, phylum, and kingdom, until it finds a reliable taxonomic container. That higher-level prediction is then mapped to a representative species associated with the predicted node.

Confidence-aware fallback decision rule

Low-confidence species predictions are redirected to a taxonomically safer representative label.

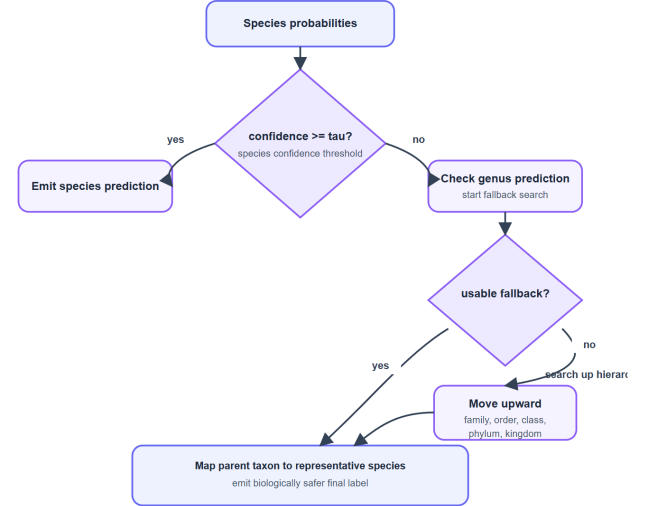


Figure 8: Confidence-aware fallback rule.

This strategy is motivated by the competition metric: a conservative taxonomically nearby answer is usually preferable to a low-confidence species guess that lands in a distant branch. In practice, the fallback threshold is set to 0.7 for species confidence. At the same time, the strongest reported leaderboard result is still driven primarily by the taxonomic-loss model itself, so we treat fallback as an auxiliary decision rule rather than the main source of gain. The rule is closely related to prior work on accuracy-specificity trade-offs, confidence calibration, and selective prediction under uncertainty (Deng et al. 2012; Guo et al. 2017; Geifman and El-Yaniv 2019).

Figure 8 summarizes the confidence-aware fallback rule used at inference time.

4.5 Training procedure and hyperparameters

Training uses stratified 80/10/10 splits to preserve species frequencies across train, validation, and evaluation sets. All images are resized to 224 x 224 and normalized with ImageNet statistics. Training augmentation includes random resized crops with scale range 0.7 to 1.0, horizontal flips, rotations up to 15 degrees, and color jitter. The data pipeline relies on pre-cropped multi-scale ROI images so that training time is spent on model learning rather than repeated image extraction.

Optimization uses AdamW with a base learning rate of 3e-4, weight decay of 0.05, mixed-precision training, and gradient accumulation over two steps (Loshchilov and Hutter 2019). Backbone parameters are fine-tuned more conservatively through a

reduced learning rate multiplier, while the fusion layers and heads train at the base rate. A cosine annealing schedule with warm restarts controls the learning rate over a 50-epoch budget (Loshchilov and Hutter 2017), and early stopping monitors `val_tax_score` rather than ordinary validation loss because that quantity better matches the competition objective.

5 Experiments and Results

5.1 Experimental setup

Experiments were implemented with PyTorch Lightning and `timm` (Falcon and the PyTorch Lightning Team n.d.; Wightman n.d.), together with a modular command-line training pipeline. We compared single-scale and multi-scale variants, evaluated the benefit of taxonomic loss, and examined several design alternatives including attention-based fusion inspired by transformer-style sequence modeling (Vaswani et al. 2017), DINOv2 backbones (Oquab et al. 2023), and independent hierarchical heads. Model selection emphasized the validation taxonomic score rather than plain cross-entropy because the leaderboard metric values biologically close mistakes more than exact top-1 accuracy.

The experimental workflow also prioritized reproducibility. Configurations were stored in YAML, data preparation was scripted, checkpoints were selected automatically from validation performance, and inference scripts generated submission files directly from the trained models.

The runs also saved TensorBoard event logs for the four-scale experiments, which means training curves for loss and validation taxonomic score are available as raw logging artifacts even though the output directory does not contain a separately exported curve figure.

Table 1 summarizes the principal training configuration used for the multi-scale ConvNeXtV2 experiments.

5.2 Quantitative results

Table 2 summarizes the main leaderboard results. The ROI-only baseline establishes a starting point, after which multi-scale context gives a large improvement. Adding wider scales improves the private score from 2.65 to 2.12, and including full-image context reduces it further to 2.05. Taxonomic loss adds another gain, culminating in the best private leaderboard score of 1.94 for the 4-scale

Setting	Value
Data split	Stratified 80/10/10 by species
Input size	224 x 224
Scales	1x, 3x, 5x, optionally full image
Backbone	ConvNeXtV2-Base
Fusion	Concatenation + Layer-Norm/Linear/GELU/Dropout
Batch size	16
Epoch budget	50
Optimizer	AdamW
Learning rates	3e-4 for new layers and 3e-5 for backbones
Scheduler	Cosine annealing with warm restarts
Precision	Mixed precision (16-mixed)
Gradient accumulation	2 steps
Early stopping	Patience 10 on validation taxonomic score

Table 1: Principal training configuration for the multi-scale ConvNeXtV2 experiments.

Model variant	Public LB	Private LB
ROI-only ConvNeXtV2 baseline	2.77	2.65
3-scale multi-scale model	2.11	2.12
4-scale multi-scale model	2.18	2.05
3-scale + taxonomic loss	2.27	2.00
Best model: 4-scale + taxonomic loss	2.65	1.94
1st-place reference solution	1.65	1.44

Table 2: Public and private leaderboard scores for the main model variants.

model.

To make the improvements easier to interpret, Table 3 reports the absolute and relative gains on the private leaderboard with respect to the ROI-only baseline and between nearby ablations.

As shown in Table 3, these numbers support several additional findings. First, the dominant source of improvement is the move from a tight ROI-only classifier to a multi-scale representation; the initial addition of context gives a much larger gain than any later architectural refinement. Second, the jump from three scales to four scales is real but modest, indicating diminishing returns once immediate and medium-range context are already available. Third, the taxonomic-distance objective contributes a fairly consistent extra improvement of about five to six percent on top of the corresponding pure cross-entropy models, which strengthens the claim that objective alignment matters independently of encoder capacity.

The public and private leaderboard behavior is

Comparison	Private delta	Relative gain
ROI-only $\rightarrow$ 3-scale	-0.53	20.0%
ROI-only $\rightarrow$ 4-scale	-0.60	22.6%
3-scale $\rightarrow$ 4-scale	-0.07	3.3%
3-scale $\rightarrow$ 3-scale + taxonomic loss	-0.12	5.7%
4-scale $\rightarrow$ 4-scale + taxonomic loss	-0.11	5.4%
ROI-only $\rightarrow$ best model	-0.71	26.8%

Table 3: Absolute and relative private-leaderboard gains for the main ablation comparisons.

also informative. The best 4-scale taxonomic-loss model attains the strongest private score despite not having the best public score, while the ROI-only and plain multi-scale variants show much smaller public-private gaps. This suggests that hierarchy-aware training may generalize better to the hidden test distribution even when the public subset is somewhat noisy or not fully representative.

Another useful quantitative takeaway is that the strongest final model does not close the entire gap to the top-ranked solution, but it narrows it substantially. Relative to the 1st-place private score of 1.44, the final 1.94 system removes most of the baseline error while still leaving room for more expressive fusion, better calibration, or stronger pretrained encoders.

The saved TensorBoard logs also allow me to reconstruct representative training curves for the two four-scale runs. Figure 9 is a four-panel comparison between the plain four-scale model and the four-scale taxonomic-loss variant. The top-left panel shows training loss, the top-right panel shows validation loss, the bottom-left panel shows validation species accuracy, and the bottom-right panel shows validation taxonomic score.

Several trends are visible in these curves. First, both runs descend rapidly during the early phase of optimization, which indicates that pretrained ConvNeXtV2 backbones adapt quickly to the marine ROI setting. Second, the taxonomic-loss run achieves clearly lower validation loss in its recorded window and steadily improves its validation taxonomic score from roughly 0.82 to 0.53, which is consistent with the intended effect of directly optimizing a hierarchy-aware objective. Third, species accuracy rises quickly for both models and then enters a flatter regime, suggesting that later improvements come more from making biologically closer mistakes than from large gains in ordinary top-1 accuracy. Finally, only the taxonomic-loss run logs `val_tax_score`, so the bottom-right panel naturally contains a single curve.

Taxonomic level	Internal evaluation accuracy
Kingdom	1.0000
Phylum	0.9682
Class	0.9530
Order	0.9193
Family	0.8959
Genus	0.8878
Species	0.8774

Table 4: Internal evaluation accuracy at each taxonomic level.

Saved internal evaluation outputs provide a complementary view of model behavior across the hierarchy. Table 4 shows that accuracy degrades smoothly as predictions move from coarse to fine taxonomic levels, which is exactly the pattern expected from a representation that preserves high-level biological structure even when exact species boundaries remain challenging.

The main takeaway from this hierarchy-wise breakdown in Table 4 is that most errors are not catastrophic. Accuracy drops by only 1.81 percentage points from family to genus and by another 1.04 points from genus to species, which implies that a substantial fraction of the residual species mistakes occur among very nearby relatives. By the time performance reaches the species level, the model has already resolved the broad branch of the taxonomy correctly for a large majority of examples, which is precisely the kind of behavior encouraged by the competition metric.

Figure 10 visualizes this pattern at the family level, where the strongest mass remains concentrated on the diagonal.

Figure 11 shows the same evaluation view at the species level, where the task becomes visibly more fine-grained.

5.3 Qualitative analysis and ablations

The experimental trajectory provides a coherent picture of what mattered most.

- Adding 3x context gave the strongest context gain, while 5x and full-image scales produced smaller but still useful increments.
- Taxonomic loss improved robustness by moving probability mass away from distant species and toward closer relatives.
- Attention-based fusion was explored, but simple concatenation remained the preferred choice because it was more stable and delivered comparable accuracy at lower memory and compute

Figure 5. Training dynamics of the four-scale models

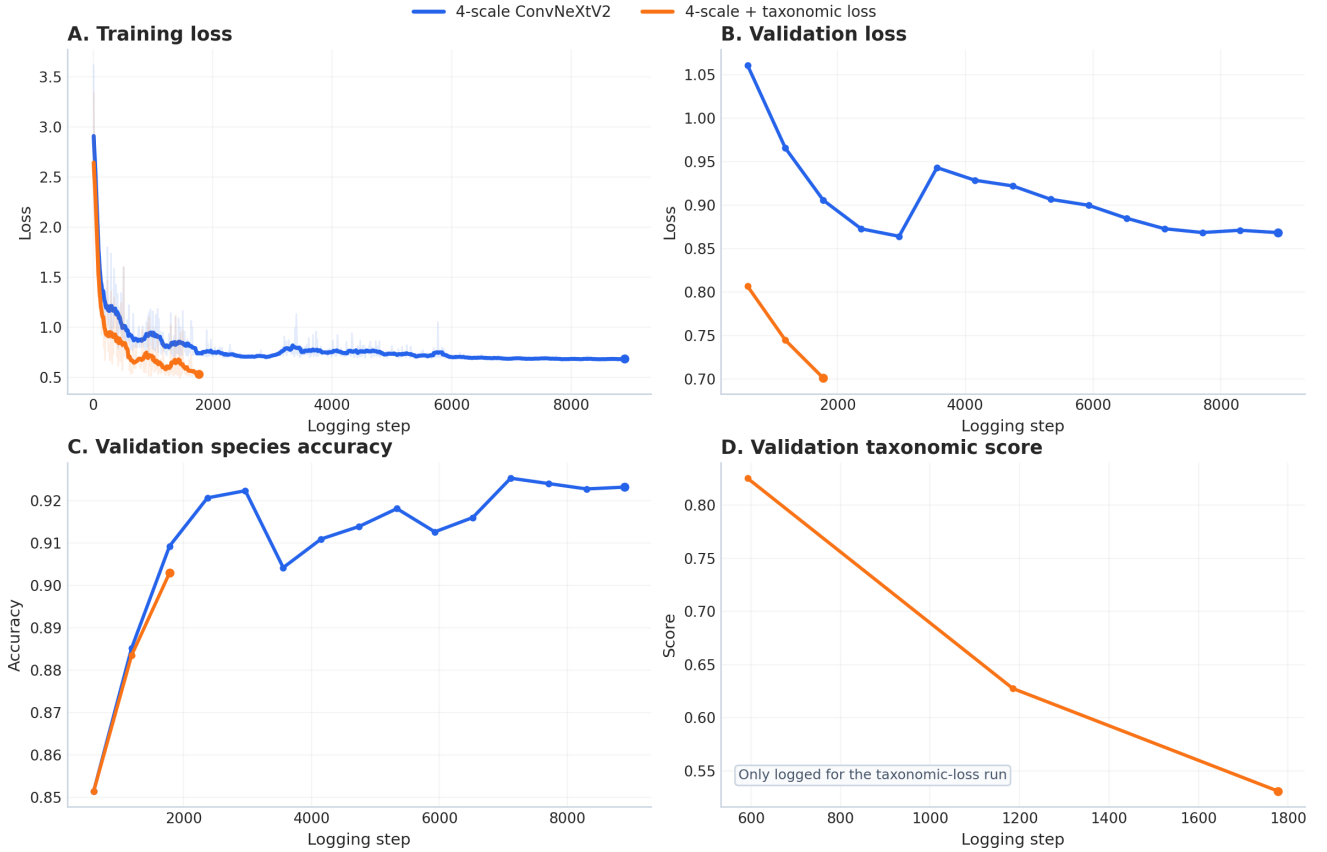


Figure 9: Training curves for the four-scale models.

cost (Vaswani et al. 2017).

- Later ablations showed that independent hierarchical heads can reduce error propagation from high taxonomic ranks, but they also weaken the structural constraints that make sequential heads biologically coherent.

Qualitatively, the model behaves most sensibly when the organism is ambiguous but the surrounding scene is informative. Context scales help distinguish visually similar species that occupy different habitats, while hierarchy-aware prediction reduces the most damaging long-range errors when the species prediction is uncertain.

The saved internal reports also reveal where the remaining mistakes concentrate. At the species level, coarse or visually ambiguous labels such as *Caridea*, *Hexactinellida*, *Porifera*, *Sebastes*, and *Psathyrometra fragilis* show noticeably lower recall than the easiest leaf categories. This pattern is consistent with the confusion matrices: errors accumulate among morphologically similar taxa, sponge-like organisms with weak local texture cues, and classes whose visual identity depends strongly on context rather than on the tight ROI alone.

Another useful observation is that family and species confusion matrices remain mostly diagonal rather than fragmenting into large off-diagonal blocks. This suggests that the model’s typical failures are local substitutions within a biologically related neighborhood, not arbitrary jumps across distant phyla. From the standpoint of the competition metric, that is a favorable error profile even when exact top-1 species accuracy is imperfect.

6 Discussion and Limitations

The strongest result of this work is not simply that a larger model performs better, but that alignment between representation, supervision, and evaluation matters. The competition score rewards biologically reasonable uncertainty, and the proposed system explicitly supports that behavior through multi-scale context, hierarchical heads, taxonomic distance-aware loss, and auxiliary hierarchy-aware fallback. These components work together: multi-scale crops make the representation more informative, the hierarchy organizes the prediction problem, and the loss makes the species probabilities better match the scoring rule. The internal evaluation staircase from perfect kingdom accuracy down to 87.74 per-

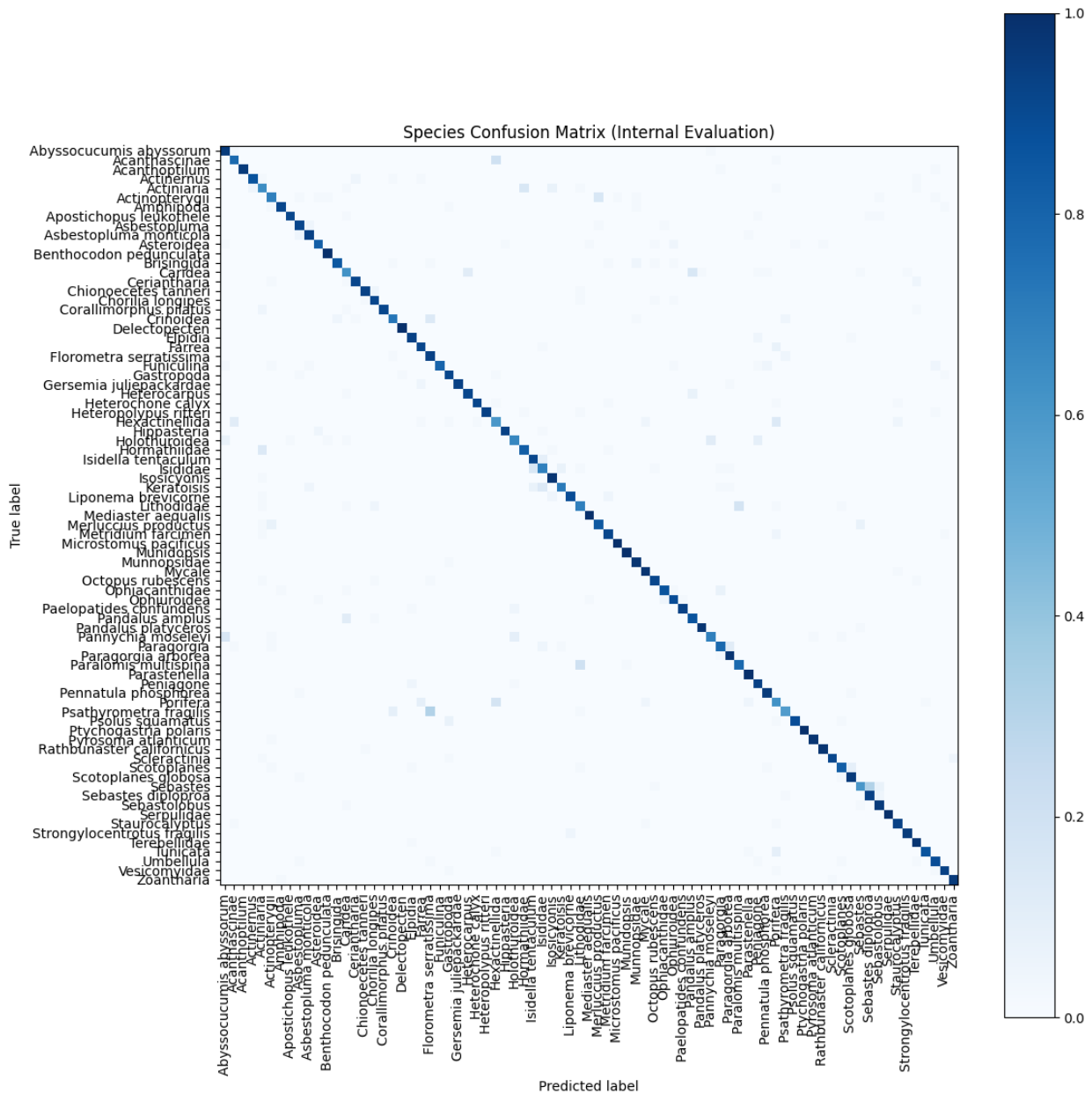


Figure 11: Species-level confusion matrix.

cent species accuracy reinforces this interpretation: even when the model misses the leaf label, it often remains on the correct high-level branch of the tree.

At the same time, the approach has several limitations. The multi-scale architecture is computationally heavy, especially once full-image context is added. The hierarchical cascade can also propagate mistakes from upper ranks into lower ones, which motivated later experiments with independent heads. A fallback rule can reduce catastrophic errors, but it can also collapse rare cases into more common representatives of the predicted higher-level taxon. The internal reports further show that the hardest remaining errors are concentrated in fine-grained, visually similar marine taxa where even family-level cues can be subtle. Finally, the gap between public and private scores indicates that local validation and leaderboard behavior are not perfectly aligned, so reported gains must be interpreted with some caution.

7 Conclusion and Future Work

This report presented a taxonomy-aware multi-scale classifier for FathomNet 2025. Starting from a ConvNeXt-based baseline, we showed that adding broader context windows substantially improves marine species recognition, and that replacing a flat species loss with a taxonomic distance-aware objective improves alignment with the competition metric. The final 4-scale model achieved a private leaderboard score of 1.94, making hierarchy-aware training and context fusion the two most important ingredients of the system.

Several future directions are promising. A more selective fusion mechanism could retain the benefits of context while reducing the large computational footprint of independent multi-scale encoders; the attention-based variants explored in this project provide a natural starting point. Better uncertainty calibration could make the fallback rule more principled, for example by predicting non-leaf taxa directly when species evidence is weak rather than mapping them back to representative leaves. Finally, lightweight ensembles or distillation from stronger transformer-based teachers may close more of the remaining gap to the top leaderboard solutions without making the deployment pipeline prohibitively expensive.

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